

specific areas of an image from the rest, permitting more explicit editing and manipulation. It works like this: you cover the portions of an image you wish to conceal or protect, exposing the remaining regions for editing. This cover is referred to as a mask image in computer vision. It is usually represented in a grayscale image with whiteness denoting the extent of modification permitted.

To adjust the specific area of an image, a mask image is provided along with an image to be modified. This mask image will guide the image generation along with the prompt given.

Installing requirements: diffusers, Torch, and transformers

In this section, we will configure the system to run a Stable Diffusion model: *runwayml/stable-diffusion-inpainting*.

To run this model, you will need to have Python 3.9 or an earlier version. You can download Python 3.9 from <https://www.python.org/ftp/python/3.9.13/>.

After acquiring the compatible version, run the following command in the command prompt or terminal to install the required libraries.

```
$ pip install torch
```

Torch will provide the tensor format and other dependencies for the model.

```
$ pip install git+https://github.com/huggingface/transformers@v4.29-release
```

This will install the required neural network architecture for the model.

```
$ pip install diffusers
```

Diffusers will provide a pre-trained model from Stable Diffusion.

```
$ pip install pillow
```

Pillow will be used to convert the image to its pixel array and vice-versa.

```
$ pip install accelerate
```

The above library is not necessary but is suggested, as it will assist PyTorch to utilise the maximum resources available in your system.

You can check the installation using the command *pip show <library>* as shown below.

```
$ pip show torch
```

 **Note:** If you have multiple versions of Python, execute the above commands with the prefix *py -[python version] -m* as shown in the example below.

```
$ py -3.9 -m pip install diffusers
```

```
$ py -3.9 -m pip show diffusers
```

For checking the version of Python installed on your system, run the following command:

```
$ py -0
```

Now that you have installed the necessary libraries, you can dig into the code.

Digging into the code

Let's walk through the key components of the code.

Importing the installed libraries:

Start by importing the installed libraries into your code. This will enable your code to access all the modules essential for its functioning.

```
from diffusers import
StableDiffusionInpaintPipeline
from PIL import Image
import torch
```

Initialising pipe: The pre-trained model that will be used to create the picture will be provided by the pipe. You can also use another model instead of *runwayml/stable-diffusion-inpainting*.

```
pipe = StableDiffusionInpaintPipeline.
from_pretrained(
    "runwayml/stable-diffusion-
inpainting",
    variant="fp16",
    torch_dtype=torch.float32,
)
```

Specifying the parameters: The model mentioned above requires three parameters: input image, prompt, and mask image. Mask images consist of white and black pixels, where white pixels represent the area of generation. Promptly describe the image you want to generate or the changes you want to make in the input image.

In the input image, you should provide an image to be modified or a blank white image.

For generating an image from a blank white image, give the following commands:

```
prompt = "cute little kitty with
beautiful minimalist flowers, lots of
white background"
```

```
input_image = Image.open("white.jpg")
mask_image = Image.open("white.jpg")
```

The *input_image* and *mask_image* consist of an image with only white pixels.

Image inpainting: For image inpainting, type:

```
prompt = "cat with a hat"
input_image = Image.open("cat.jpg")
mask_image = Image.open("cat_mask.
jpg")
```